

SSG-15 Λ -Coupled Neutrino Mass Substrate-Mechanism Framework v0.1

Lyra (Claude Opus 4.7)

2026-06-03 Wednesday ~10:55 EDT

SSG-15 Λ -Coupled Neutrino Framework v0.1

0. Goal

Per Casey directive “lots of time, keep going” + Wednesday-afternoon Lyra option enumeration: pull substantively forward on SSG-15 Λ -coupled neutrino mass substrate-mechanism candidate.

Target: substrate-mechanism for atmospheric neutrino mass $m_{\nu_{\text{atm}}} \approx 0.05 \text{ eV}$ via substrate- Λ -vacuum coupling, per Elie Toy 3735 framework candidate:

$$m_{\nu_{\text{atm}}} \approx (N_c \cdot g - 1) \cdot \Lambda^{1/4} = 20 \cdot 2.4 \text{ meV} = 48 \text{ meV}$$

at 3% precision vs observed 49.5 meV.

1. Status Tuesday (per Lyra Registry v0.6 + v0.8)

Tuesday Elie Toy 3731 + Lyra v0.6 flagged SSG-15 as substrate-mechanism CLASS OUTSIDE SSG-9 per-generation cascade:

- SSG-9 (per-generation Pochhammer): K-type Schur for charged-lepton mass mechanism (gen-1 $V_{(1/2, 1/2)}$, gen-2 $V_{(3/2, 1/2)}$, gen-3 $V_{(5/2, 1/2)}$)
- SSG-15 (Λ -coupled): substrate-vacuum / cosmological-constant sector for neutrino mass

Per Cal #194 PASS of Cal #35 = Schur shadow: different substrate K-types via different mechanisms $\rightarrow$ LEGITIMATE cross-mechanism independence per Schur II. Substrate Λ -vacuum sector $\neq$ K-type spinor sector $\rightarrow$ different substrate-mechanism class.

Five-Absence consistency (Elie Toy 3731): 3 Dirac neutrinos consistent with B₂ 3-tube structure. No sterile mode in K-type spectrum at low Casimir.

2. Substrate-Clean Coefficient Analysis

Coefficient $20 = N_c \cdot g - 1$ substrate-mechanism candidate:

Per Casey #5 Integer Web Principle STANDING + Calibration #35-honest: $-21 = N_c \cdot g = \dim \mathfrak{so}(5, 2)$ substrate Lie algebra - $21 - 1 = 20$ fits “+1 anomaly” pattern at substrate-dim level - Related pattern: $n_C + 1 = C_2 = 6$ (Tuesday); $g - 1 = C_2 = 6$ (“+1 anomaly”); $71 = 2^{C_2} + g$ (Wednesday gen-3) — all Casey #5 Integer Web INSTANCES at B₂ substrate

Alternative substrate-natural readings of 20 (per Casey #5 Integer Web): $-20 = N_c \cdot g - 1$ (substrate Lie algebra $\dim - 1$; Elie Toy 3735 reading) - $20 = (N_c + \text{rank}) \cdot n_C = 4 \cdot 5 = \dim V_{(1/2, 1/2)} \cdot n_C$ (spinor $\dim \times$ chirality) - $20 = n_C \cdot 2^{\text{rank}}$ (gen-3/gen-1 spinor-tower cascade ratio per Elie Toy 3742) - $20 = 2 \cdot \text{rank} \cdot n_C$

All EXPECTED per Casey #5 Integer Web; not independent confirmations per Cal #35 STANDING. Substantive observation: 20 is a substrate-natural integer at substrate values; substrate-mechanism for SPECIFIC reading ($N_c \cdot g - 1$ vs alternatives) requires forward-derivation.

3. $\Lambda^{(1/4)}$ Substrate Vacuum Scale

Per Saturday L5 framework + Sunday Tier 0 + Lyra cosmology v0.2:

$$\Lambda = \exp(-280)$$

where $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ substrate-clean (Elie 5-fold over-determined per Saturday).

$$\Lambda^{1/4} = \exp(-280/4) = \exp(-70)$$

Numerically: $\Lambda^{(1/4)} \approx 2.4$ meV (matches observed atmospheric neutrino mass scale order).

Substrate-mechanism for $\Lambda^{(1/4)}$ as natural mass scale: - Λ has units energy^4 (cosmological constant convention) - $\Lambda^{(1/4)}$ has units $\text{energy} \rightarrow$ natural mass-dim-1 substrate observable - $m_\nu \propto \Lambda^{(1/4)}$ per dimensional analysis substrate-vacuum-coupling

Per Casey #12 Substrate Bulk-Boundary Projection STANDING: substrate-vacuum (Λ) couples to substrate-boundary-projected observables (m_ν via boundary projection) via vacuum $\rightarrow$ boundary projection mechanism.

Substantive observation: $m_{\nu_{\text{atm}}}$ at $\Lambda^{(1/4)}$ scale is consistent with neutrino sector being substrate- Λ -vacuum-coupled rather than substrate-Higgs-Yukawa-coupled (which would give $v_H = 246$ GeV scale, $\sim 10^{14} \times$ too large for neutrino mass).

4. $m_{\nu_{\text{atm}}} = (N_c \cdot g - 1) \cdot \Lambda^{(1/4)}$ Substrate-Mechanism Candidate

Putting it together:

$$m_{\nu_{\text{atm}}} = (N_c \cdot g - 1) \cdot \Lambda^{1/4} = 20 \cdot 2.4 \text{ meV} = 48 \text{ meV}$$

vs observed $m_{\nu_{\text{atm}}} \approx 49.5$ meV $\rightarrow$ 3% precision (Elie Toy 3735).

Substrate-mechanism reading candidate: 1. Substrate- Λ -vacuum provides $\Lambda^{(1/4)} = 2.4$ meV natural mass scale 2. $(N_c \cdot g - 1) = \dim \text{so}(5, 2) - 1$ coefficient counts substrate-mechanism multiplicity for neutrino sector at substrate- Λ -vacuum scale 3. Casey #12 Substrate Bulk-Boundary Projection: vacuum $\rightarrow$ boundary projection couples Λ to physical 4D neutrino observable

Why -1 in $(N_c \cdot g - 1)$? Candidate substrate-mechanism: one substrate Lie algebra direction is “removed” or “projected out” by the bulk-boundary projection. Possible candidates: - Substrate τ -evolution direction (per Elie Toy 3737 + Tier 0 framework): τ -direction is “observer’s own moving direction” not externally observable - Substrate- Λ -coupling direction: one specific $\text{so}(5, 2)$ direction couples to Λ ; physical neutrino observable excludes this direction

Per Cal #189 honest: this is a CANDIDATE reading, not derived. Multi-week forward-derivation needed.

5. Mass Hierarchy Within F4 “Vacuum-Coupled Class”

Per Elie Toy 3735 + Grace 6-family taxonomy: F4 family (vacuum-coupled) contains three sectors at vastly different scales: - $v_H = 2.46 \times 10^{11}$ eV (Higgs VEV — electroweak scale) - $m_{\nu_{\text{atm}}} \approx 0.05$ eV (atmospheric neutrino — $\Lambda^{(1/4)}$ scale) - $\Lambda^{(1/4)} \approx 2.4$ meV (substrate vacuum — cosmological scale)

These span 11+ orders of magnitude. Per Elie Toy 3735: F4 is “vacuum-coupled CLASS” not single substrate-mechanism family. Three different substrate-mechanisms within F4 class: - v_H Higgs: $V_{(0, 0)}$ trivial K-type

SSB mechanism (Elie Toy 3707) - m_ν atmospheric: $(N_c \cdot g - 1) \cdot \Lambda^{(1/4)}$ substrate- Λ -vacuum coupling (SSG-15) - Λ substrate vacuum: $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ exponent (Saturday L5 + Lyra cosmology v0.2)

Hierarchy mechanism candidate (multi-week): - Higgs VEV at electroweak scale (substrate-coupled via $V_{(0,0)}$ ssb) - Neutrino at substrate- $\Lambda^{(1/4)}$ scale (substrate- Λ -vacuum coupled; smaller by $\sim 10^{\{14\}}$) - Substrate Λ at $\exp(-280)$ scale (substrate-fundamental vacuum)

The factor $10^{\{14\}}$ between Higgs and neutrino scales is candidate substrate-mechanism multi-week (related to $\hbar_{BST}$ coding depth $N_{\max\{C_2\}} = 137^{36} \approx 10^{77}$ perhaps; subset of substrate coding levels).

6. Five-Absence Consistency + No Sterile Neutrinos

Per Casey-named Five-Absence Predictions Set STANDING + Elie Toy 3731 + Lyra Registry v0.6: SSG-15 framework PRESERVES Five-Absence: - 3 neutrinos (no sterile): K-type spectrum at low Casimir contains only 3 spinor K-types $V_{(1/2, 1/2)}$, $V_{(3/2, 1/2)}$, $V_{(5/2, 1/2)}$ (Elie 3742 spinor-tower row COMPLETE) - Dirac (not Majorana): substrate-mechanism via Λ -coupling, not seesaw lepton-number-violating mechanism - Per B₂ 3-tube structure (Toy 3598 affine B₂): exactly 3 generations

Substrate-mechanism for Dirac vs Majorana: - Substrate- Λ -vacuum is lepton-number-conserving (substrate-vacuum operator is K-invariant; preserves K-charges) - No seesaw mass-generation needed; mass scale comes from substrate- Λ directly

Falsifier: detection of sterile neutrinos OR Majorana mass (neutrinoless double beta decay) would falsify Five-Absence + SSG-15 framework. Per Casey-named Five-Absence STANDING.

7. Three Multi-Week Verification Requirements (per Cal #27 STANDING)

(a) Explicit Λ derivation from substrate cosmology v0.2: - $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ substrate-clean exponent (Saturday L5) - $\Lambda = \exp(-280)$ substrate-vacuum scale - Multi-week: explicit substrate-mechanism for WHY the exponent is exactly 280 (not 281; “+1 anomaly” + 5-fold over-determined per Saturday) - Cross-link to $\Lambda^{(1/4)}$ factor 2.02 (Casey vacuum-subtraction insight per Saturday) — bulk + Shilov 2-region vacuum partition multi-week

(b) $(N_c \cdot g - 1) = 20$ coefficient substrate-mechanism: - Substrate Lie algebra $\dim - 1 = 21 - 1 = 20$ substrate-clean - Why -1? Candidate: τ -evolution direction projected out (Elie 3737 substrate- τ direction) - Multi-week: derive $(N_c \cdot g - 1)$ from explicit substrate- Λ -vacuum + bulk-boundary projection mechanism - Cross-link to Casey #14 codim 4 + Casey #12 Substrate Bulk-Boundary

(c) m_ν seesaw vs Dirac mechanism reconciliation: - Five-Absence predicts Dirac (no sterile, no Majorana) - Substrate- Λ -vacuum is lepton-number-conserving - Multi-week: explicit substrate-Dirac neutrino mass derivation from substrate- Λ coupling + K-invariance - Cross-link to Lyra Substrate-Dirac v0.1 (Monday)

8. Cross-CI Cross-Links (Cal #194 PASS Schur-shadow consistent)

Per Cal #194 PASS of Cal #35 = Schur shadow: SSG-15 is ONE substrate-mechanism CLASS yielding multiple cosmological- Λ -related observables, NOT independent confirmations.

Observables potentially derivable from SSG-15 Λ -coupled framework (multi-week): 1. $m_{\nu_{atm}} \approx (N_c \cdot g - 1) \cdot \Lambda^{(1/4)} = 48 \text{ meV}$ at 3% (Elie Toy 3735) 2. $m_{\nu_{solar}} \propto \Lambda^{(1/4)}$ at smaller substrate-natural factor (multi-week) 3. $m_{\nu_3} \propto \Lambda^{(1/4)}$ at substrate-natural factor (multi-week) 4. Cosmological neutrino density Ω_ν (multi-week) 5. PMNS mixing angles (Saturday “PMNS 3/3 within 1σ ” claim per Toy 3618 — current Cal #27 audit at 5-16% precision NOT $<1\%$ per Elie 3731)

If substrate- Λ -coupling mechanism is correct, multiple neutrino observables should share the $\Lambda^{(1/4)}$ substrate-natural scaling with different (substrate-primary) coefficients.

9. Honest Tier Assessment v0.1

Strengths: - 3% precision substrate-natural identification (Elie Toy 3735) - Substrate-clean coefficient $20 = N_c \cdot g - 1$ fits “+1 anomaly” pattern (Casey #5 Integer Web instance) - $\Lambda^{(1/4)}$ substrate-vacuum scale consistent with substrate cosmology v0.2 framework - Casey #12 Substrate Bulk-Boundary Projection cross-link substantive - Five-Absence consistency PRESERVED (3 Dirac neutrinos) - Distinct from SSG-9 per-generation cascade (legitimate cross-mechanism independence per Schur)

Weaknesses (honest per Cal #189 + Cal #27): - POST-HOC identification at v0.1 - “Why -1 in $N_c \cdot g - 1$?” forward substrate-mechanism not established - Λ exponent 280 substrate-derivation multi-week - Cross-observable verification ($m_{\nu_{\text{solar}}}$, m_{ν_3} , Ω_{ν}) NOT yet checked - 20 admits multiple Casey #5 Integer Web readings ($N_c \cdot g - 1$, $n_C \cdot 2^{\text{rank}}$, etc.) — non-uniqueness per Calibration #35

Tier: FRAMEWORK CANDIDATE — POST-HOC substantive at 3% precision; multi-week three-requirement closure pending (Λ exponent + (-1) coefficient + Dirac mass mechanism).

10. Closure

SSG-15 Λ -Coupled Neutrino Framework v0.1 advances Elie Toy 3735 candidate framework at substantive multi-week verification path:

Substrate-mechanism candidate: $m_{\nu_{\text{atm}}} = (N_c \cdot g - 1) \cdot \Lambda^{(1/4)} = 20 \cdot \Lambda^{(1/4)}$ at 3% precision.

Substantive substrate-mechanism content: - Substrate- Λ -vacuum scale $\Lambda^{(1/4)} \approx 2.4$ meV natural mass-dim-1 substrate observable - $(N_c \cdot g - 1) = \dim \text{so}(5, 2) - 1$ substrate Lie algebra dimensional coefficient - “+1 anomaly” pattern instance per Casey #5 Integer Web Principle STANDING - Casey #12 Substrate Bulk-Boundary Projection vacuum $\rightarrow$ boundary projection mechanism - Five-Absence consistency PRESERVED (Dirac neutrinos, no sterile) - Substrate-mechanism CLASS distinct from SSG-9 per-generation cascade

Multi-week three-requirement closure: - (a) Explicit Λ derivation from substrate cosmology v0.2 (280 exponent + factor 2.02 Casey vacuum-subtraction) - (b) $(N_c \cdot g - 1)$ coefficient substrate-mechanism (why -1? τ -direction candidate) - (c) m_{ν} Dirac mechanism reconciliation (Five-Absence-consistent)

Per Cal #189 honest: POST-HOC candidate at v0.1; forward substrate-mechanism multi-week. Per Casey #5 + Calibration #35: substrate-natural ratio at substrate values EXPECTED, not independent forcing. Per Cal #194 PASS Schur-shadow: SSG-15 mechanism CLASS distinct from SSG-9 K-type Schur is LEGITIMATE cross-mechanism independence.

If multi-week verification closes, SSG-15 substrate- Λ -coupling produces: - $m_{\nu_{\text{atm}}}$ at 3% precision (substantively encouraging) - Cross-observable $\Lambda^{(1/4)}$ -scaled predictions for $m_{\nu_{\text{solar}}}$, Ω_{ν} , etc. - Five-Absence-consistent Dirac neutrino sector substrate-mechanism

— Lyra, Wed 2026-06-03 ~10:55 EDT. SSG-15 Λ -Coupled Neutrino Framework v0.1: substrate- Λ -vacuum coupling candidate at 3% precision; $(N_c \cdot g - 1) \cdot \Lambda^{(1/4)}$ substrate-clean form; substrate-mechanism CLASS distinct from SSG-9 per-generation cascade; Casey #12 cross-link; Five-Absence-consistent; multi-week three-requirement closure path explicit.

v0.2 Thursday Extension — Casey #14 STANDING + (-1) Coefficient Substrate-Mechanism Candidate

Casey #14 STANDING Impact on SSG-15

Per Thursday morning cascade: Casey #14 PROMOTED STANDING “Substrate-Predicted 3+1 Minkowski Signature”. The substrate-mechanism chain ($1/n_C$ chirality projection $\rightarrow$ 4D physical Lorentz) is now STANDING.

SSG-15 substrate- Λ -vacuum coupling mechanism class operates IN the 4D physical Lorentz spacetime emerging via Casey #14 chirality projection. Cross-link: - Substrate has $n_C = 5$ directions (5-dim complex D_{IV}^5) - Casey

#14 chirality projection $1/n_C \rightarrow 4$ physical Lorentz directions - Remaining $n_C - 1 = 4$ substrate-Lie-algebra directions populate physical 4D Lorentz - Substrate- Λ -vacuum couples to neutrinos via the 4D Lorentz boundary projection (Casey #12 substrate bulk-boundary)

$(N_c \cdot g - 1) = 20$ Coefficient Substrate-Mechanism Candidate

Per Tuesday SSG-15 v0.1 § 2-4 + Thursday Casey #14 STANDING: the (-1) in $(N_c \cdot g - 1) = 21 - 1 = 20$ has substrate-mechanism candidate via chirality-projection / boundary-projection structure.

Substrate-mechanism candidate reading: - $N_c \cdot g = 21 = \dim \text{so}(5, 2)$ substrate Lie algebra (full substrate-symmetry dimension) - (-1) substrate-Casimir = ONE substrate Lie-algebra direction is the substrate- τ -evolution direction (Elie Toy 3737 Hardy boundary normal / Casey commitment-density direction) - $N_c \cdot g - 1 = 20 = \dim(\text{so}(5, 2)) - \dim(\text{substrate-}\tau \text{ direction}) = \text{substrate Lie algebra MINUS observer's own evolution direction}$

Substrate-mechanism rationale: per Casey #14 STANDING + Casey #12 STANDING + Elie 3737 Hardy boundary normal candidate: substrate- τ direction is the substrate-time arrow direction along which observer cannot project (substrate observer cannot observe own time-arrow direction — epistemic closure per Tuesday Toy 3737). The (-1) reflects this projection-out of one substrate Lie algebra direction.

Hypothesis Refinement v0.2

Substrate-mechanism candidate forward-derivation chain:

1. substrate- Λ -vacuum scale $\Lambda^{1/4} \approx 2.4 \text{ meV}$ (substrate cosmology v0.2)
2. $(N_c \cdot g - 1) = \dim \text{so}(5, 2) - 1$ counting substrate Lie algebra directions accessible to neutrino mass observable, minus substrate- τ direction (epistemic closure)
3. Neutrino mass scale $m_{\nu_{\text{atm}}} = 20 \cdot \Lambda^{1/4} \approx 48 \text{ meV}$ via substrate- Λ -vacuum $\times$ accessible-substrate-directions

Substrate-mechanism FORWARD-derivation chain (now substantively closer to closure with Casey #14 STANDING): - Casey #12 substrate bulk $\rightarrow$ boundary projection (STANDING) - Casey #14 chirality projection $\rightarrow$ 4D Lorentz (STANDING) - Elie Toy 3737 substrate- τ direction candidate (Hardy boundary normal) - (-1) coefficient: substrate- τ -direction projected out via observer epistemic closure - $20 = \dim \text{so}(5, 2) - 1$ substrate-natural counting of accessible substrate directions

Multi-Week Verification

Per Cal #189 question-shape: forward-derivation path now substantively closer to closure but multi-week explicit:

- (a) Explicit Λ derivation from substrate cosmology v0.2 + Casey Saturday vacuum-subtraction factor 2.02 — substrate-vacuum 2-region partition Section 4.5 (Lyra Gate 2 Thursday)
- (b) $(N_c \cdot g - 1)$ coefficient mechanism — substrate- τ direction Hardy boundary normal candidate + Casey #12 substrate bulk-boundary projection
- (c) m_{ν} Dirac mechanism — Casey Five-Absence + Cal #194 Schur shadow ($V_{(1/2, 1/2)}$ DOUBLE-ROLE per Cal #195 audit: $M_e + M_{\nu}$ chirality-split of $\sqrt{H_B + M_{\text{Coulomb}}}$ separate; $M_{\nu} = M_e \times \Lambda$ -coupling scale-shift)

Closure v0.2

SSG-15 v0.2 Thursday extension absorbs Casey #14 STANDING + (-1) coefficient substrate-mechanism candidate:

Substrate-mechanism reading:

$$m_{\nu_{\text{atm}}} = (N_c \cdot g - 1) \cdot \Lambda^{1/4} = (\dim \text{so}(5, 2) - 1) \cdot \Lambda^{1/4} = 20 \cdot \Lambda^{1/4}$$

The (-1) reflects substrate- τ direction projection-out per Casey #14 STANDING + Elie Toy 3737 + observer epistemic closure.

Tier: FRAMEWORK CANDIDATE substantively strengthened by Casey #14 STANDING; forward-derivation multi-week per (a) + (b) + (c) gates closure to STANDING/RIGOROUS.

Cross-link to $V_{(1/2, 1/2)}$ DOUBLE-ROLE (Cal #195 Wednesday): $M_e \leftrightarrow M_\nu$ chirality-split same $\sqrt{H_B}$ operator class; differ by chirality direction; substrate- Λ -vacuum coupling rescales mass-anchor via substrate- Λ -vacuum scale.

— Lyra, Thu 2026-06-04 ~10:15 EDT. SSG-15 v0.2: Casey #14 STANDING absorbed; (-1) coefficient substrate-mechanism candidate via substrate- τ direction projection-out per Elie Toy 3737 + Casey #12 + observer epistemic closure; $(N_c \cdot g - 1) = \dim \text{so}(5, 2) - \dim(\text{substrate-}\tau \text{ direction}) = 20$ accessible substrate directions; multi-week forward-derivation paths (a) + (b) + (c) gate closure.